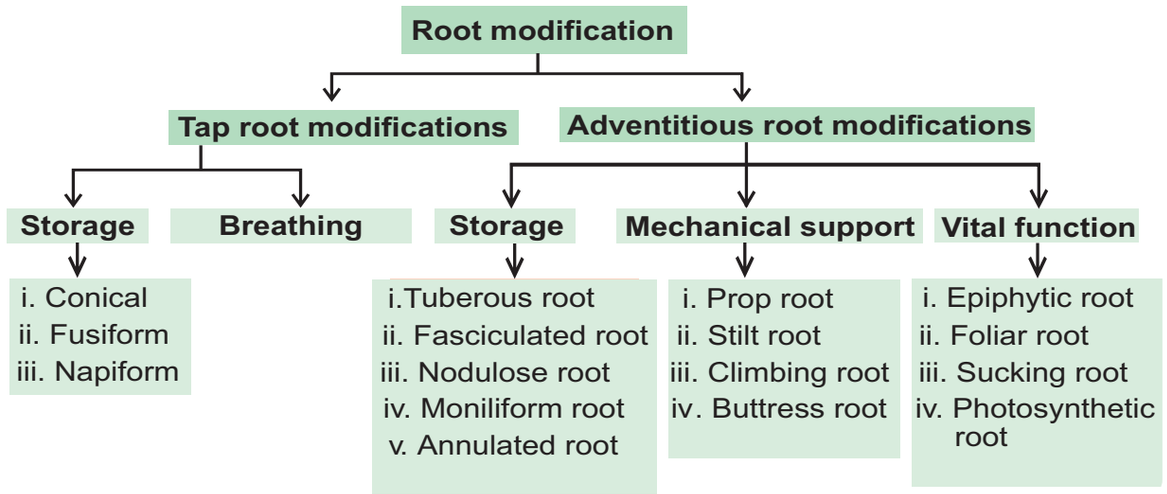


roots arise from various regions of the plant body. These are bunch of thread-like roots nearly equal in size which are collectively called **fibrous** root system generally found in grasses. Example: *Oryza sativa, Eleusine coracana*.

3.5.2 Functions of root

Root performs two kinds of functions namely primary and secondary functions.

Primary function

1. Absorb water and minerals from soil.
2. Help to anchor the plant firmly in the soil.

Secondary function

In some plants roots perform additional functions. These are called **secondary functions.** To perform additional functions, structure of roots are modified.

3.5.3 Modifications of root

I. Tap root modification

a. Storage roots

1. Conical Root

These are cone like, broad at the upper end and gradually tapering towards the lower end. Example: *Daucus carota*.

2. Fusiform Root

These roots are swollen in the middle and tapering towards both ends. Example: *Raphanus sativus*

3. Napiform Root

These roots are very broad at the upper end and suddenly tapers like a tail at the lower end. Example: *Beta vulgaris*

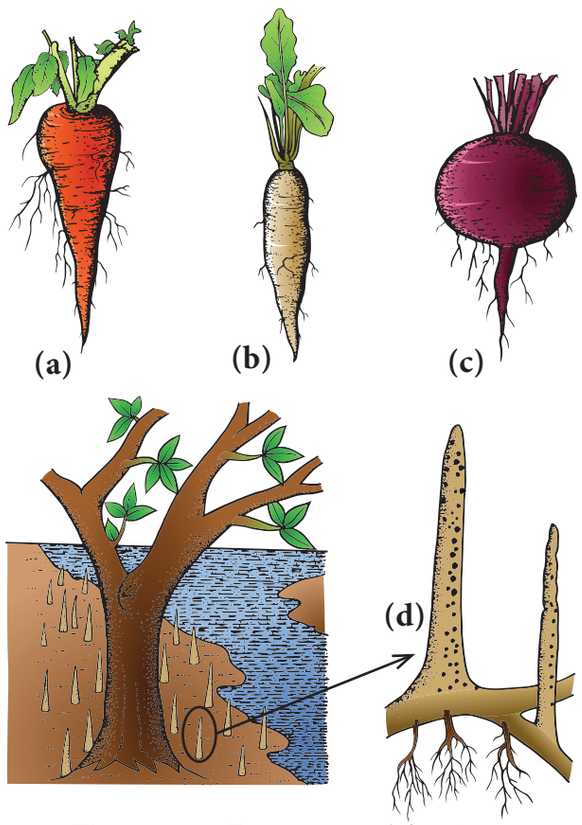


Figure 3.4: Tap root modifications
(a) *Daucus carota* **(b)** *Raphanus sativus*
(c) *Beta vulgaris* **(d)** *Avicennia* - pneumatophores

b. Breathing root

Some mangrove plants like ***Avicennia, Rhizophora, Bruguiera*** develop special kinds of roots (Negatively geotropic) for respiration because the soil becomes saturated with water and aeration is very poor. They have a number of breathing pores on pneumatophores for exchange of gases.

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